BHOOMIFY

PROJECT REPORT

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CERTIFICATE

This is to certify that this project report entitled "**BHOOMIFY**" is a bonafide record of work done by **ANEYA ALIN (B25CS1214), ARATHY KRISHNA A.M (B25CS1217), ESHAN M.S (B25CS1226), GOKUL VISWA P (B25CS1228), JESSAE GEJO KUNNEL (B25CS1233), NEHA ROY (B25CS1241), S.R NIREEKSH (B25CS1250), SALMAN SABITH (B25CS1252), SARIN S JOHN (B25CS1253)** of the first semester Computer Science Engineering Branch towards the partial fulfilment of the requirements for the award of Bachelor of Technology Degree in Computer Science Engineering of the APJ Abdul Kalam Technological University, Kerala.

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ABSTRACT

Environmental degradation and climate change have emerged as major global concerns due to unsustainable human activities, excessive consumption of resources, and lack of environmental accountability. Educational institutions play a vital role in shaping environmentally responsible behaviour among students. However, the absence of structured incentive mechanisms often results in low participation in green initiatives.

The **"Bhoomify"** project is designed as a digital solution to promote environmental responsibility by introducing a structured points-based incentive system. Students are awarded points for participating in environmentally beneficial activities, which can later be redeemed for rewards. The system integrates a mobile application with a backend service and database to track participation, encourage sustained engagement, and promote awareness of sustainable practices.

The project demonstrates how technology can be effectively utilized to support environmental initiatives by combining software engineering principles with sustainability objectives. By integrating digital record-keeping, reward mechanisms, and transparent ranking systems, the project encourages responsible behaviour while reinforcing concepts of sustainability and environmental stewardship.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Rapid industrialization, urbanization, and lifestyle changes have significantly increased environmental stress in recent decades. Excessive waste generation, resource depletion, and irresponsible consumption patterns have contributed to climate change, pollution, and ecological imbalance. While awareness of environmental issues has increased, active participation in sustainability initiatives remains limited, particularly among students.

One effective method to encourage sustainable behaviour is through incentive-based motivation. When individuals are rewarded for environmentally responsible actions, participation levels increase and positive habits are reinforced. With the widespread availability of smartphones and digital platforms, technology offers an efficient medium to implement such incentive systems.

The “Bhoomify” project aims to bridge this gap by providing a digital platform that tracks environmentally responsible activities, assigns points, and encourages participation through rewards and recognition.

1.2 Need for the Project

Despite environmental education being part of academic curricula, many students lack practical exposure to sustainability initiatives. Common issues include:

- Lack of motivation to participate in green activities.
- Absence of transparent tracking mechanisms.
- Difficulty in monitoring student participation.
- Limited reinforcement of eco-friendly behaviour.

A digital incentive system addresses these challenges by providing measurable outcomes, transparency, and motivation. The “Bhoomify” project fulfils this requirement by integrating environmental awareness with modern application technology.

1.3 Objectives of the Project

The main objectives of the “Bhoomify” project are:

- To design a digital platform that encourages eco-friendly behaviour among students.
- To assign incentive points for participation in environmental activities.
- To maintain transparent records of points earned and redeemed.
- To promote sustainable habits through reward-based motivation.
- To demonstrate the application of software technology in environmental studies.
- To enhance student engagement through digital tracking and leaderboards.

1.4 Scope of the Project

The scope of the “Bhoomify” project includes:

- Student registration and role-based access.
- Tracking of environmental participation through points.
- Maintenance of point history and redemption records.
- Display of leaderboards to encourage healthy competition.
- Reward redemption through a digital shop.

The project is designed as an academic and technical demonstration suitable for educational environments.

CHAPTER 2

LITERATURE REVIEW

2.1 Technology in Environmental Management

Studies indicate that digital platforms play an increasingly important role in environmental monitoring, awareness, and participation. Mobile applications have been successfully used to track waste management, carbon footprints, and sustainability metrics.

2.2 Incentive-Based Sustainability Models

Research highlights that incentive-based systems significantly increase participation in environmentally responsible activities. Reward mechanisms encourage behavioural change by reinforcing positive actions.

2.3 Role of Gamification in Education

Gamification techniques such as point systems, rankings, and rewards have been proven to improve student engagement and motivation. Integrating these concepts with environmental education enhances participation and learning outcomes.

CHAPTER 3

METHODOLOGY

3.1 System Design Approach

The project follows a structured methodology consisting of frontend development, backend service implementation, and database integration. A client–server architecture was adopted to ensure modularity and clarity.

3.2 Frontend Development

A mobile application was developed using Flutter to provide a user-friendly interface for students and faculty. The application allows users to view points, history, leaderboards, and rewards.

3.3 Backend Implementation

The backend was implemented using Flask, providing RESTful APIs to handle requests such as point assignment, history retrieval, and leaderboard computation. The backend acts as an intermediary between the frontend and the database.

3.4 Database Management

A PostgreSQL database was used to store student records, point history, redemption details, and shop inventory. Structured tables ensure data consistency and traceability.

CHAPTER 4
SUSTAINABLE DEVELOPMENT GOALS (SDGs)

4.1 Introduction

The Sustainable Development Goals (SDGs), adopted by the United Nations in 2015, aim to address critical global challenges such as environmental degradation, climate change, resource depletion, and lack of sustainability awareness. Educational institutions play a crucial role in promoting sustainable development by shaping responsible attitudes and behaviours among students

The Bhoomify project – A Green Incentive and Reward Management System addresses these challenges by introducing a technology-driven incentive mechanism that motivates students to actively participate in environmentally responsible activities. By combining digital tracking, reward-based motivation, and transparency, the project aligns with multiple Sustainable Development Goals and contributes to long-term sustainability outcomes.

4.2 Alignment with Specific SDGs

- SDG 4: Quality Education

  - Acts as an educational tool that integrates sustainability concepts with real-world technological applications.
  - Enhances experiential and practical learning through hands-on participation in environmental activities.
  - Develops awareness of:
    - Environmental responsibility
    - Sustainable development practices
    - Digital systems in environmental management
    - Promotes lifelong learning and sustainability awareness through continuous engagement.

- SDG 8: Decent Work and Economic Growth
  - Introduces students to green skills such as sustainability management, digital tracking systems, and environmental accountability.
  - Encourages innovation in technology-driven sustainability solutions.
  - Lays the foundation for future green entrepreneurship and employment opportunities.
  - Supports sustainable economic thinking by linking incentives with responsible behaviour.

- SDG 11: Sustainable Cities and Communities
  - Encourages community participation within educational institutions through collective green initiatives.
  - Promotes cleaner and more sustainable campus environments by motivating responsible behaviour.

- Supports the development of sustainable communities through student-driven environmental engagement.
- Reinforces civic responsibility and environmental stewardship among young individuals.

- SDG 12: Responsible Consumption and Production
  - Promotes responsible consumption patterns by encouraging students to participate in eco-friendly and sustainable activities.
  - Encourages reduction of waste and resource misuse through awareness-driven incentive mechanisms.
  - Supports sustainable lifestyle choices by rewarding environmentally responsible behaviour.
  - Enhances understanding of efficient resource utilization through digital monitoring and participation tracking.

- SDG 13: Climate Action
  - Encourages active participation in climate-positive actions such as waste management, energy conservation, and green initiatives.
  - Indirectly contributes to reduction of carbon footprint by motivating environmentally responsible habits.
  - Raises climate awareness among students through consistent engagement and reward-based reinforcement.
  - Supports institutional efforts toward climate mitigation and sustainability awareness.

4.3 Broader Impact

- Environmental Impact
  - Encourages reduction in environmentally harmful practices through incentive-based motivation.
  - Promotes long-term adoption of sustainable habits among students.
  - Supports conservation of natural resources through increased environmental awareness.

- Social Impact
  - Encourages active student participation in sustainability initiatives.
  - Builds a sense of environmental responsibility and accountability.
  - Inspires positive behavioural change toward eco-friendly practices through rewards and recognition.

- Educational Impact
  - Strengthens the integration of technology with environmental education.
  - Enhances student understanding of sustainability concepts through digital platforms.
  - Encourages interdisciplinary learning combining software engineering and environmental studies.

CHAPTER 5

WORKING, IMPLEMENTATION AND ENGINEERING CONSIDERATIONS

5.1 Project Overview

This project presents the design and implementation of a Green Incentive and Reward Management System, developed as part of the Environmental Studies (EVS) curriculum. The primary objective of the system is to encourage environmentally responsible behaviour among students by assigning points for eco-friendly activities and allowing these points to be redeemed for rewards. The system demonstrates the practical application of software engineering concepts such as modular design, client–server architecture, data persistence, and user role management, while aligning with sustainability-focused goals.

5.2 System Architecture

The application follows a three-tier client–server architecture:

- Frontend Layer – A mobile application developed using Flutter.
- Backend Layer – A RESTful web service implemented using Flask.
- Database Layer – A PostgreSQL database managed through Supa-base.

This separation ensures clarity of responsibility across system components and improves maintainability and scalability within the project scope. All interactions between the frontend and backend occur through HTTP-based JSON APIs, ensuring standardized and platform-independent communication.

5.3 Role-Based System Design

The system supports two primary user roles:

- Faculty
  - Add and manage student records.
  - Assign points for environmentally beneficial activities.
  - Manage reward items in the green shop.
  - View point allocation and redemption history.

- Students
  - View current point balance.
  - Track detailed point history.
  - View leaderboard rankings.
  - Redeem rewards using accumulated points.

Role separation ensures controlled access to system functionality and reflects real-world institutional workflows.

5.4 Backend Working

The backend is implemented using Flask, providing a lightweight yet effective REST API layer. The backend is intentionally designed to be stateless, meaning each request is processed independently without storing session information.

The backend handles:

- Authentication and role verification.
- Student record management.
- Point assignment and ledger maintenance.
- Reward redemption validation.
- Dynamic leaderboard generation.
- Leaderboard Implementation.

The leaderboard is generated dynamically by aggregating student point data and sorting it in descending order. This avoids redundant storage and ensures that rankings always reflect the latest data, demonstrating an efficient and normalized data design approach.

5.5 Database Design

The database schema is structured to ensure clarity, consistency, and traceability. Key tables include:

- Student details
- Point history (ledger-style recording of earned and spent points)
- Shop inventory
- Redemption history

Historical data is preserved to allow transparency and auditability of all actions performed within the system.

5.6 Frontend Working

The frontend is developed using Flutter, enabling a responsive and user-friendly mobile interface. The application dynamically updates its UI based on data received from the backend. All network communication is centralized through a dedicated service layer, ensuring consistency and reducing duplication. The interface provides clear feedback through loading indicators, empty states, and error messages, enhancing usability.

5.7 Engineering Decisions and Rationale

Several design decisions were taken deliberately to align with project scope and educational objectives:

- Computed leaderboard instead of stored rankings to avoid data redundancy.
- Stateless backend for simplicity and reliability.
- RESTful APIs for clarity and interoperability.
- Centralized API service layer for maintainability.

These decisions ensure the system remains easy to understand while demonstrating correct architectural practices.

5.8 Testing and Validation

The system was validated through a combination of:

- Manual API testing using browser-based tools.
- End-to-end testing via the Flutter application.
- Verification of data consistency across frontend, backend, and database.

All major features were tested successfully, including login, point assignment, reward redemption, and leaderboard display.

5.9 Deployment

The backend service is deployed on a cloud-based hosting platform, while the database is hosted using Supa-base. This demonstrates the use of modern cloud infrastructure and managed services within the scope of an academic project. Environment configuration is handled through variables to allow flexibility and easy redeployment.

5.10 Team Contribution

This project was completed as a team-based academic exercise, with responsibilities distributed across system design, implementation, testing, and documentation. Collaborative planning and iterative development were followed to ensure timely completion of all project requirements.

5.11 Environmental Relevance

The system aligns with the objectives of Environmental Studies by promoting sustainable practices through positive reinforcement. By incentivizing environmentally responsible actions, the project demonstrates how technology can be leveraged to support awareness, participation, and behavioural change in sustainability initiatives.

5.12 Scope and Limitations

The project is developed as a technical demonstration suitable for academic evaluation. Advanced features such as large-scale concurrency handling, advanced security mechanisms, and performance optimization are considered beyond the current scope.

5.13 Conclusion

The project successfully demonstrates the implementation of a complete full-stack application with clear architectural separation, role-based access, and sustainability-focused objectives. It effectively integrates frontend, backend, and database components while remaining aligned with the learning outcomes of the EVS curriculum.

Fig 5.1 App Logo

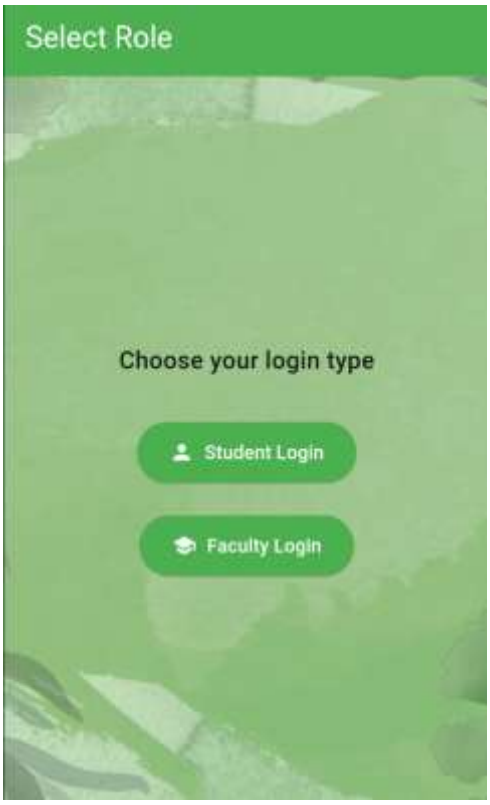

Fig 5.2 Role Selector

Fig 5.3 Student Dashboard

Fig 5.4 Faculty Dashboard

CHAPTER 6
RESULTS AND DISCUSSION

6.1 System Functionality

The system successfully performs all intended operations, including student management, point assignment, reward redemption, and leaderboard display.

6.2 Environmental Impact

By incentivizing eco-friendly activities, the project encourages students to actively participate in sustainability initiatives. The transparent tracking mechanism promotes accountability and awareness.

6.3 Learning Outcomes

The project enhanced understanding of both environmental responsibility and software system design. Students gained exposure to practical applications of sustainability concepts using technology.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Bhoomify project successfully demonstrates how digital technology can be utilized to promote environmental awareness and responsible behaviour. Through a structured incentive system, the project encourages student participation in sustainability initiatives while reinforcing environmental education objectives.

7.2 Future Scope

The project can be extended to include:

- Integration with real-world environmental activities.
- Expansion to community-level participation.
- Advanced analytics for sustainability tracking.
- Integration with institutional environmental programs.
- Awarding an “Eco-Champion of the Year” certificate to students with the highest green credit points.

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